ESP32 ↔ SD CARD MODULE — UNPOWERED CONTINUITY / SHORT-CIRCUIT TEST SHEET

⚠ ALL TESTS MUST BE PERFORMED WITH USB POWER DISCONNECTED.

ESP32 and SD module must be completely UNPOWERED for every test. USB cable DISCONNECTED. All 12 tests are resistance (Ω) tests only.

Multimeter setup: Mode = Resistance (Ω) | Range = 200 Ω | Black probe → COM | Red probe → VΩmA

Date	03-10-2026	Multimeter	DT-830B
ESP32 board		Meter range	200 Ω
SD module			

Fill in: Actual Reading, Pass/Fail, Notes (and the summary).

Test #	Test Description	Red Probe	Black Probe	USB Connected?	Expected Reading	Actual Reading	Pass/Fail	Notes
1	MISO continuity	SD Module MISO	ESP32 GPIO19	NO	~0–2 Ω	1.4	✓	
2	MISO to GND	SD Module MISO	SD Module GND	NO	1 / OL	1	✓	
3	MISO to 3.3V	SD Module MISO	SD Module 3.3V	NO	1 / OL	1	✓	
4	CS continuity	SD Module CS	ESP32 GPIO5	NO	~0–2 Ω	1.2–1.3	✓	
5	CS to GND	SD Module CS	SD Module GND	NO	1 / OL	1	✓	
6	CS to 3.3V	SD Module CS	SD Module 3.3V	NO	1 / OL	1	✓	
7	MOSI continuity	SD Module MOSI	ESP32 GPIO23	NO	~0–2 Ω	1.2	✓	
8	MOSI to GND	SD Module MOSI	SD Module GND	NO	1 / OL	1	✓	
9	MOSI to 3.3V	SD Module MOSI	SD Module 3.3V	NO	1 / OL	1	✓	
10	SCK continuity	SD Module SCK	ESP32 GPIO18	NO	~0–2 Ω	1.2–1.3	✓	
11	SCK to GND	SD Module SCK	SD Module GND	NO	1 / OL	1	✓	
12	SCK to 3.3V	SD Module SCK	SD Module 3.3V	NO	1 / OL	1	✓	

SUMMARY

MISO result	Pass	
CS result	Pass	
MOSI result	Pass	

SCK result	Pass	
Shorts detected (any GND / 3.3V test FAIL)	NO	
General notes		

Result cells and Shorts detected calculate automatically from the Pass/Fail column; overwrite manually if needed.